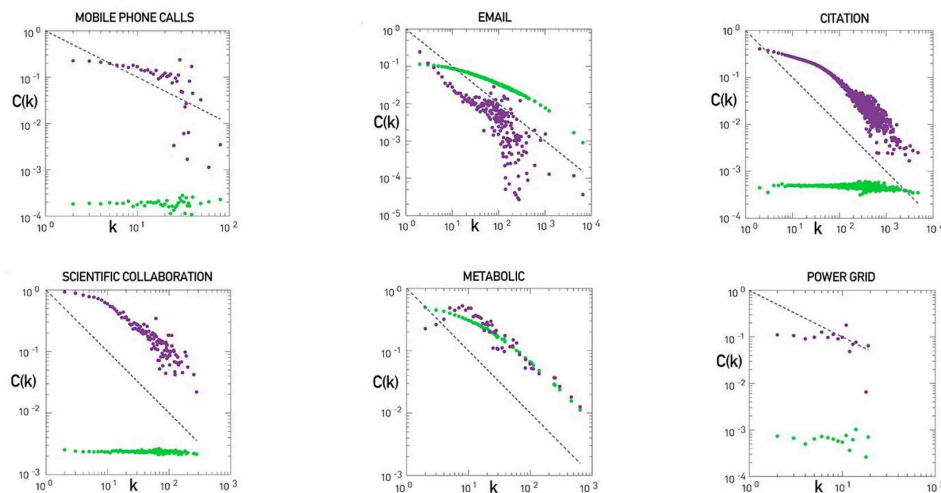


L7: Clustering Coefficient Versus Degree in Practice



Hierarchy in Real Networks. (Image 9.36) from networksciencebook.com

Do we see this inversely proportional relationship between the clustering coefficient and degree in real-world networks? And if so, is it a reliable indicator of hierarchical modularity?

The visualization shows plots of the node clustering coefficient $C(k)$ versus node degree k for six real-world networks (*purple dots*). You can find additional information about these six datasets in your textbook.

The green dots refer to a corresponding degree-preserving randomized network – this is an important step of the analysis because by randomizing the network connections (*without changing the degree distribution*), we remove the community structure of the original network from the randomized network.

The black dotted line corresponds to the relation $C(k) = 1/k$. As we saw in the previous page, this is the slope of $C(k)$ versus k in the deterministic hierarchically modular network that we saw earlier.

Note that some networks, such as the network of scientific collaborations or the citation network show clearly an inverse relation between $C(k)$ and k , especially for nodes with degree $k > 10$. Further, when we randomize the connections of those networks, removing the community structure, $C(k)$ becomes independent of k . So, we expect that such networks exhibit a hierarchically modular structure.

The situation is not always so clear, however. Some other networks, such as the metabolic network or the email network, show an inverse relation between $C(k)$ and k – but the same is true for the corresponding randomized networks! This raises the concern that the inverse relation between $C(k)$ and k may be caused in some cases at least by the degree distribution – and not by the community structure of the network.

Finally, there are also networks in which we do not see a decreasing relation between $C(k)$ and k , such as the mobile phone calls network or the power grid.

To summarize, hierarchical modularity is an important property of many (but not all) real-world networks, and when present, it exhibits itself with an inversely proportional between the clustering coefficient and the node degree. However, this relation is not sufficient evidence for the presence of hierarchical modularity.